

Unitary Coupled Cluster theory on Quantum Computers with Parametric Matrix Models

Patrick Cook and Danny Jammooa

*Facility for Rare Isotope Beams and Department of Physics and Astronomy,
Michigan State University, East Lansing, Michigan 48824, USA*

Morten Hjorth-Jensen

Department of Physics and Center for Computing in Science Education, University of Oslo, N-0316 Oslo, Norway

Dean Lee

*Facility for Rare Isotope Beams and Department of Physics and Astronomy,
Michigan State University, East Lansing, Michigan 48824, USA*

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I. INTRODUCTION

Structure of article

1. Introduction with motivation
2. Discussion of the Lipkin model
3. Setting up time evolution
4. PMMs and time evolution
5. Trotterization and CC theory

Quantum computers offer a promising route to simulate quantum many-body systems by preparing variational wavefunctions and measuring their energies. The Variational Quantum Eigensolver (VQE) is a leading hybrid algorithm that finds ground-state energies by classically optimizing the parameters of a quantum circuit. A crucial ingredient in VQE is the choice of ansatz – the form of the trial wavefunction. The Unitary Coupled Cluster (UCC) ansatz has emerged as a physically motivated choice, inspired by the success of classical coupled-cluster theory in quantum chemistry. Unlike hardware-agnostic ansätze, UCC builds in knowledge of the system’s excitation structure, making it easier to converge and often more accurate for a given circuit depth.

In this work, we develop the formalism of the UCC approach for quantum computing, including a full derivation of the UCC wavefunction form and the use of the Baker–Campbell–Hausdorff (BCH) expansion to analyze the Trotterization error. We illustrate the method using the nuclear pairing Hamiltonian as an example – a simple yet nontrivial many-body model relevant for nuclear structure physics. Nuclear pairing is a well-studied model where identical nucleons form Cooper pair-like bound states in discrete energy levels. It presents a challenging strongly correlated problem that, despite its simplicity, requires beyond mean-field methods for accurate solutions. Recent studies have begun applying VQE and adaptive methods to pairing models, making it an ideal testbed for UCC on quantum hardware.

II. MATHEMATICAL FRAMEWORK AND ENTANGLEMENT MEASURES

A. Hamiltonian and Symmetry.

The Lipkin–Meshkov–Glick (LMG) model describes N spin- $\frac{1}{2}$ particles interacting through collective spin operators. For four fermions, the case we consider in the examples here, each level has degeneracy $d = 4$, leading to different total spin values. The two levels have quantum numbers $\sigma = \pm 1$, with the upper level having $2\sigma = +1$ and energy $\varepsilon_1 = \varepsilon/2$. The lower level has $2\sigma = -1$ and energy $\varepsilon_2 = -\varepsilon/2$. That is, the lowest single-particle level has negative spin projection (or spin down), while the upper level has spin up. In addition, the substates of each level are characterized by the quantum numbers $p = 1, 2, 3, 4$.

The Hamiltonian of the system is given by

$$\hat{H} = \hat{H}_0 + \hat{H}_1 + \hat{H}_2$$

$$\hat{H}_0 = \frac{1}{2}\varepsilon \sum_{\sigma,p} \sigma a_{\sigma,p}^\dagger a_{\sigma,p}$$

$$\hat{H}_1 = \frac{1}{2}V \sum_{\sigma,p,p'} a_{\sigma,p}^\dagger a_{\sigma,p'}^\dagger a_{-\sigma,p'} a_{-\sigma,p}$$

$$\hat{H}_2 = \frac{1}{2}W \sum_{\sigma,p,p'} a_{\sigma,p}^\dagger a_{-\sigma,p'}^\dagger a_{\sigma,p'} a_{-\sigma,p}$$

where V and W are constants. The operator H_1 can move pairs of fermions while H_2 is a spin-exchange term. The latter moves a pair of fermions from a state $(p\sigma, p' - \sigma)$ to a state $(p - \sigma, p'\sigma)$.

We are going to rewrite the above Hamiltonian in

terms of so-called quasispin operators

$$\begin{aligned}\hat{J}_+ &= \sum_p a_{p+}^\dagger a_{p-} \\ \hat{J}_- &= \sum_p a_{p-}^\dagger a_{p+} \\ \hat{J}_z &= \frac{1}{2} \sum_{p\sigma} \sigma a_{p\sigma}^\dagger a_{p\sigma} \\ \hat{J}^2 &= J_+ J_- + J_z^2 - J_z\end{aligned}$$

and the number operator

$$\hat{N} = \sum_{p\sigma} a_{p\sigma}^\dagger a_{p\sigma}.$$

We will now rewrite the Hamiltonian in terms of the above quasi-spin operators and the number operator

$$N = \sum_{p,\sigma} a_{p\sigma}^\dagger a_{p\sigma}. \text{label} : N \quad (1)$$

Going through each term of the Hamiltonian and using the expressions for the quasi-spin operators we obtain

$$H_0 = \varepsilon J_z, \quad (2)$$

$$H_1 = \frac{1}{2} V (J_+^2 + J_-^2), \quad (3)$$

and

$$H_2 = \frac{1}{2} W (-N + J_+ J_- + J_- J_+). \quad (4)$$

The above expressions can in turn be used to show that the Hamiltonian commutes with the various quasi-spin operators. This leads to quantum numbers which are conserved. Let us first show that $[H, J^2] = 0$, which means that J is a so-called *good* quantum number and that the total spin is a conserved quantum number.

The matrix elements are given by $\langle J, J_z | H | J', J'_z \rangle$. The Hamiltonian is hermitian and we obtain after all this labor of ours

$$H_{J=2} = \begin{bmatrix} -2\varepsilon & 0 & \sqrt{6}V & 0 & 0 \\ 0 & -\varepsilon + 3W & 0 & 3V & 0 \\ \sqrt{6}V & 0 & 4W & 0 & \sqrt{6}V \\ 0 & 3V & 0 & \varepsilon + 3W & 0 \\ 0 & 0 & \sqrt{6}V & 0 & 2\varepsilon \end{bmatrix} \text{label} : HJ = 2 \quad (5)$$

III. PARAMETRIC MATRIX MODELS

A. Scientific Background and Motivation

The ability to model, predict, and control the behavior of complex quantum systems is central to the ongoing second quantum revolution. Across quantum chemistry, condensed matter, and quantum technology, the

challenge lies in bridging the gap between microscopic Hamiltonian descriptions and emergent macroscopic phenomena. Traditional methods—such as exact diagonalization or coupled-cluster theory—are computationally demanding, often scaling exponentially with system size. Machine learning (ML) offers a complementary route by enabling data-driven discovery of patterns and effective models. However, standard ML approaches typically lack interpretability, stability, and physical consistency.

A Parametric Matrix Model describes a system (or dataset) through a matrix whose entries depend on a set of tunable parameters. Formally, one writes

$$M(\theta) = \sum_i \theta_i A_i,$$

where A_i are fixed basis matrices and θ_i are parameters to be learned or optimized. Such models can represent Hamiltonians, covariance matrices, kernels, or linear transformations whose structure reflects known physics, symmetries, or constraints.

Unlike dense neural networks that learn arbitrary mappings, PMMs impose a low-dimensional, interpretable structure. This makes them ideal for

- learning models with few but physically meaningful parameters,
- ensuring stability and symmetry preservation (e.g., Hermiticity, orthogonality),
- reducing overfitting and improving generalization.

Because $M(\theta)$ is parametric and smooth in its parameters, it can be integrated seamlessly with automatic differentiation frameworks (e.g., PyTorch, TensorFlow). This allows one to train PMMs end-to-end as part of a neural architecture—for instance, as a differentiable layer encoding physical interactions or correlations.

In quantum mechanics, the spectrum and dynamics of a system depend on its Hamiltonian matrix. PMMs allow ML models to learn maps from parameters to observables (e.g., energies, eigenvectors, correlation functions). This has direct implications for:

- learning effective Hamiltonians,
- building surrogate models for quantum simulations,
- predicting phase transitions or control parameters.

PMMs enable hybrid learning, where parts of the model are learned from data and parts are constrained by physical principles. For example

- embedding PMMs in a neural network can encode unitarity, time-reversal symmetry, or particle-number conservation,

- combining PMMs with variational learning yields **interpretable latent spaces** corresponding to physically relevant degrees of freedom.

The recent work [?] shows that PMMs can learn physically consistent effective theories directly from data — in particular, how parameters influence spectral and dynamical properties. This bridges machine learning and quantum many-body physics, allowing one to

- infer Hamiltonian landscapes,
- emulate complex correlated systems, and
- guide quantum control or materials design.

Parametric Matrix Models are valuable in ML because they

1. Encode structure, symmetries, and interpretability.
2. Reduce model complexity and improve generalization.
3. Provide differentiable, physics-informed representations.
4. Connect data-driven learning with analytical and physical insight.

In short — they offer a bridge between pure data fitting and mechanistic understanding, making them ideal for scientific machine learning and quantum technology applications.

IV. COUPLED CLUSTER THEORY (REVIEW)

In classical many-body quantum chemistry, the *Coupled Cluster* (CC) method uses an exponential ansatz for the wavefunction. One writes

$$|\Psi_{CC}\rangle = e^{\hat{T}} |\Phi_0\rangle, \quad (6)$$

where $|\Phi_0\rangle$ is a reference (usually Hartree–Fock) determinant, and the cluster operator $\hat{T}$ is a sum of excitation operators

$$\hat{T} = \hat{T}_1 + \hat{T}_2 + \cdots + \hat{T}_N. \quad (7)$$

Here $\hat{T}_1$ generates all single-particle excitations from occupied to virtual orbitals, $\hat{T}_2$ generates all double excitations, and so on. For example, in second-quantized notation with creation ($a_a^\dagger$) and annihilation (a_i) operators, one may write

$$\hat{T}_1 = \sum_{i,a} t_i^a a_a^\dagger a_i, \quad \hat{T}_2 = \frac{1}{4} \sum_{i<j,a<b} t_{ij}^{ab} a_a^\dagger a_b^\dagger a_j a_i,$$

and in general T_n promotes n electrons. Truncation of the sum yields common CC models: e.g. the singles-and-doubles approximation (CCSD) retains only $T_1 + T_2$

The exponential form ensures *size-consistency* (size-extensivity) of the method: non-interacting subsystems have additive energy because $e^{\hat{T}}$ factorizes. By contrast, a truncated configuration interaction (CI) expansion lacks this property. The nonlinearity of $e^{\hat{T}}$ also effectively includes portions of higher excitations (e.g. $\hat{T}_2^2$ yields quadruple excitations) even when truncating at doubles which accelerates convergence toward the exact (FCI) solution. In practice, the CC energy is obtained by inserting the ansatz into the Schrödinger equation $H|\Psi_{CC}\rangle = E|\Psi_{CC}\rangle$ and projecting against excited determinants. Equivalently, one works with the non-Hermitian similarity-transformed Hamiltonian $\bar{H} = e^{-\hat{T}} H e^{\hat{T}}$, solving

$$\langle \Phi_0 | \bar{H} | \Phi_0 \rangle = E_{CC}, \quad \langle \Phi_\mu | \bar{H} | \Phi_0 \rangle = 0,$$

for all excited determinants $|\Phi_\mu\rangle$ up to the truncation level. Because $e^{\hat{T}}$ is not unitary, the CC energy is not obtained via a variational minimization (indeed conventional CC is not variational. In fact, CCSD scales as $O(N^6)$ with system size (more generally, CC with up to k -fold excitations scales like N^{k+2}). Despite these costs, CCSD and its perturbatively-augmented variant CCSD(T) are among the most accurate methods for weakly-correlated molecular systems.

A. The Unitary Coupled Cluster Ansatz

The *Unitary Coupled Cluster* (UCC) approach modifies the CC ansatz to make it unitary, suitable for quantum hardware. In UCC one defines an anti-Hermitian cluster operator $\hat{A} = \hat{T} - \hat{T}^\dagger$ and writes

$$|\Psi_{UCC}\rangle = e^{\hat{A}} |\Phi_0\rangle, \quad (8)$$

so that $e^{\hat{A}}$ is manifestly unitary. For UCCSD, $\hat{A}$ includes the usual singles and doubles excitation terms, but each excitation amplitude t_i^a is paired with its de-excitation to form $(t_i^a - t_i^{a*})$ in the exponent. This ansatz preserves the size-extensivity of CC but ensures unitarity, making it a natural variational form for quantum circuits.

In principle the UCC energy

$$E = \langle \Phi_0 | e^{-(\hat{A} - \hat{A}^\dagger)} H e^{\hat{A} - \hat{A}^\dagger} | \Phi_0 \rangle$$

could be variationally minimized. However, classically evaluating this expectation is intractable for large systems due to the non-terminating Baker–Campbell–Hausdorff expansion. Thus UCC is rarely used in classical CC codes. On a quantum computer, the unitary exponentials can be directly implemented, sidestepping the need for the non-Hermitian similarity transform

Key differences from classical CC include: UCC uses a unitary operator and hence can be optimized by energy minimization, but the operators $\hat{\tau}_i - \hat{\tau}_i^\dagger$ do not commute

in general. The resulting UCC ansatz (e.g. UCCSD) thus defines a manifold of wavefunctions different from conventional CCSD. In practice one often constructs UCC ansätze by decomposing $e^{\hat{T}-\hat{T}^\dagger}$ into implementable quantum gates (via Trotterization; see below). In applications, UCCSD has been extensively studied in quantum chemistry (as a variational wavefunction and as a benchmark)

Its advantages on quantum devices include the exact preservation of norm and the ability to define variational parameters directly as gate angles, at the cost of deeper circuits.

B. Trotterization of the UCC Operator

The unitary operator $U_{\text{UCC}} = e^{\hat{T}-\hat{T}^\dagger}$ generally cannot be implemented in one piece on a quantum circuit, since $\hat{T} - \hat{T}^\dagger = \sum_i (\hat{\tau}_i - \hat{\tau}_i^\dagger)$ is a sum of non-commuting terms (each $\hat{\tau}_i$ being a fermionic excitation generator). Instead, one applies the *Trotter-Suzuki* (product-formula) decomposition to approximate the exponential of a sum by a product of exponentials of individual terms. Formally, for a Trotter step size $1/t$, one writes

$$e^{\sum_i A_i} = \lim_{t \rightarrow \infty} \left(\prod_i e^{A_i/t} \right)^t.$$

In first-order (Lie) Trotter approximation ($t = 1$), one simply takes the product once:

$$e^{\hat{T}-\hat{T}^\dagger} \approx \prod_i \exp(\theta_i (\hat{\tau}_i - \hat{\tau}_i^\dagger)), \quad (9)$$

where θ_i are the corresponding cluster amplitudes. More generally, with t steps one has

$$e^{\hat{T}-\hat{T}^\dagger} \approx \left[\prod_i e^{(\theta_i/t)(\hat{\tau}_i - \hat{\tau}_i^\dagger)} \right]^t + O(1/t),$$

with an $O(1/t)$ error per step (i.e. $O(\Delta^2)$ error in total with step size $\Delta = 1/t$).

Physically, this means repeating a sequence of gates t times can systematically improve the accuracy, at the cost of circuit depth. In practice, one often takes $t = 1$ or a small integer in order to limit gate count.

Higher-order Suzuki formulas can reduce the error scaling. For example, the second-order symmetric Trotter decomposition splits each exponential as

$$e^{A+B} \approx e^{A/2} e^B e^{A/2}, \quad (10)$$

with error $O(\Delta^3)$.

Applied to UCCSD, one could interleave half-rotations of one excitation group, a full rotation of another, then the other half-rotation back. Generally, a k th-order Suzuki formula reduces the Trotter error but requires

more gate operations (for example, a second-order sequence of m exponentials yields roughly $2m - 1$ exponentials per layer). In the quantum chemistry VQE context, first-order (even single-step) Trotterization is commonly used since variational optimization can partly absorb the Trotter error, (By contrast, in quantum phase estimation one needs high Trotter order to faithfully simulate real time-evolution.)

Coupled Cluster (CC) Theory Recap: In classical many-body theory, the coupled cluster method expresses the correlated ground state $|\Psi\rangle$ as an exponential of an excitation operator acting on a reference state (typically the Hartree-Fock state $|\Phi_0\rangle$). In standard CC, one writes

$$|\Psi_{\text{CC}}\rangle = e^T |\Phi_0\rangle,$$

where the cluster operator T is a sum of particle-hole excitation operators of increasing rank. For example, truncated to single and double excitations (the CCSD approximation), one has

$$T = T_1 + T_2,$$

with

$$T_1 = \sum_i^{\text{occ}} \sum_a^{\text{virt}} t_i^a a_a^\dagger a_i, \quad T_2 = \frac{1}{4} \sum_{i,j}^{\text{occ}} \sum_{a,b}^{\text{virt}} t_{ij}^{ab} a_a^\dagger a_b^\dagger a_j a_i.$$

Here $a_i^\dagger, a_i$ are fermionic creation/annihilation operators; indices $i, j, \dots$ label initially occupied orbitals in $|\Phi_0\rangle$, while $a, b, \dots$ label unoccupied (virtual) orbitals. The amplitudes t_i^a, t_{ij}^{ab} are determined by projecting the Schrödinger equation onto excited configurations (resulting in non-linear CC equations). Despite its high accuracy for many systems, a drawback of classical CC is its non-unitary form – the exponential e^T is not unitary because T is not anti-Hermitian. As a result, $|\Psi_{\text{CC}}\rangle$ is not strictly a variational state (one cannot compute $\langle \Psi_{\text{CC}} | H | \Psi_{\text{CC}} \rangle$ easily), and truncating T at a finite rank does not guarantee an upper bound on the true ground energy.

UCC Ansatz: The unitary coupled cluster ansatz fixes this by including the Hermitian conjugate excitations in the exponent:

$$-\Psi_{\text{UCC}}\rangle = e^{T-T^\dagger} |\Phi_0\rangle,$$

where T is defined as in standard CC, and $T^\dagger$ contains the de-excitation operators (e.g. $T_1^\dagger = \sum_{i,a} t_i^a a_i^\dagger a_a$, etc.). By construction, $T - T^\dagger$ is anti-Hermitian (since $(T - T^\dagger)^\dagger = -(T - T^\dagger)$), so the exponential $U = \exp(T - T^\dagger)$ is a unitary operator. The parameters $t_i^a, t_{ij}^{ab}, \dots$ (often denoted θ to emphasize they will be variationally optimized angles on a quantum computer) now appear in both the excitation and de-excitation terms. For instance, a single-excitation term contributes $\exp[\theta(a_a^\dagger a_i - a_i^\dagger a_a)]$ to the unitary – essentially

a fermionic rotation between orbital i and a . Because the UCC ansatz is unitary, it can be used in a variational algorithm: the energy $E(\vec{\theta}) = \langle \Phi_0 | e^{-(T-T^\dagger)}, H, e^{T-T^\dagger} | \Phi_0 \rangle$ is guaranteed to be $\geq$ the true ground energy for any choice of parameters, and can be minimized using a quantum computer to evaluate the expectation value for given $\vec{\theta}$.

Truncation and Formal Properties: In practice, one must truncate T to a manageable level of excitations (e.g. singles and doubles, giving the UCCSD ansatz). Even with T truncated, the unitary $e^{T-T^\dagger}$ generates higher-order excitations implicitly via the Baker–Campbell–Hausdorff series. This is a key distinction from classical CC: while a truncated CCSD wavefunction strictly contains up to double excitations only, a truncated UCCSD state includes singles, doubles, and higher excitations (up to N -tuple) in its expansion, because

$$e^{T-T^\dagger} = 1 + (T - T^\dagger) + \frac{1}{2!}(T - T^\dagger)^2 + \frac{1}{3!}(T - T^\dagger)^3 + \dots$$

Products like $(T_2)(-T_2^\dagger)$ in these powers can produce four-fold excitations, etc. This feature often makes UCCSD more accurate than CCSD for the same t parameters, at the cost of a more complicated wavefunction form. However, unlike classical CC, there is no known polynomial-time classical algorithm to determine the optimal UCC amplitudes t , since evaluating the energy requires exponential effort on a classical computer. This is where quantum computing comes in: we can let a quantum processor handle the state preparation $|\Psi(\vec{\theta})\rangle = e^{T-T^\dagger}|\Phi_0\rangle$ and measurement of $\langle H \rangle$, while a classical optimizer updates the parameters. This hybrid VQE approach was demonstrated in the first experimental VQE simulation of a chemical system, using a UCCSD ansatz for H_2 and other small molecules.

In summary, the UCC ansatz provides a systematically improvable family of trial wavefunctions that are guaranteed to be normalized and variational. UCC with single and double excitations (UCCSD) is often a reasonable starting point, and one can extend to triples, etc., if needed (with rapidly increasing circuit complexity). Alternatively, adaptive schemes (ADAPT-VQE) can iteratively build up T by adding only the most important excitation operators. In this work we will focus on the fixed UCCSD ansatz adapted to the pairing Hamiltonian (which effectively becomes a pUCCD ansatz – pair cluster doubles, as explained later).

A major practical challenge in realizing $U = e^{T-T^\dagger}$ on quantum hardware is that the cluster operator $T - T^\dagger$ is a sum of many non-commuting terms. For example, T_2 consists of many operators $a_a^\dagger a_b^\dagger a_j a_i$ for various combinations, and each such term generally does not commute with others (they share some creation or annihilation operators). On a quantum computer, we can implement the exponential of a single Pauli-term efficiently (e.g. $\exp[i\theta, Z_1 Z_2]$ can be done with two CNOTs and a single-qubit rotation). But we cannot directly turn

$e^{(A+B)}$ into a circuit when A and B do not commute, except by introducing approximation. The most common strategy is to use a Trotter–Suzuki decomposition. Suppose we want $U = e^{A+B}$ for two non-commuting pieces A and B . The first-order Trotter formula states:

$$e^{A+B} \approx e^A e^B,$$

with an error of order $O(\frac{1}{2}[A, B])$ (we will quantify this shortly). More generally, one can split into r steps:

$$e^{A+B} = \left(e^{A/r} e^{B/r} \right)^r + \text{error}.$$

As $r \rightarrow \infty$ the error goes to zero. For any finite r , the formula incurs a Trotter error arising from the non-commutativity. The BCH expansion for the product of exponentials is a useful way to understand the error. If $U_A = e^A$ and $U_B = e^B$, then one can show:

$$U_A U_B = \exp \left(A + B + \frac{1}{2}[A, B] + \frac{1}{12}[A, [A, B]] - \frac{1}{12}[B, [A, B]] + \dots \right)$$

Thus, $e^A e^B$ equals e^{A+B} only if $[A, B] = 0$. In general, the extra commutator terms mean $e^A e^B \neq e^{A+B}$. The lowest-order error term is $\frac{1}{2}[A, B]$. If we use $(e^{A/r} e^{B/r})^r$, the leading error after r steps is roughly $\frac{r}{2}[A/r, B/r] = \frac{1}{2r}[A, B]$. In other words, the norm of the error scales like $\sim |[A, B]|/r$. A more rigorous bound for first-order Trotter’s operator-norm error is:

$$\|e^{A+B} - e^A e^B\| \leq \frac{1}{2} \|[A, B]\| e^{\|A\| + \|B\|},$$

and similarly for multiple steps with step size $1/r$ (the error decreases as $1/r$ for large r). Higher-order Suzuki formulas can improve the error scaling (e.g. symmetric Trotter with sequence $e^{A/2} e^B e^{A/2}$ cancels the first-order commutator term), but at the cost of more gate operations.

In the context of UCC, the unitary $U = \exp\left(\sum_k \theta_k (K_k - K_k^\dagger)\right)$ is a product of many exponentials, each term K_k being an individual excitation operator (like $a_a^\dagger a_i$ for a specific i, a). We can always Trotterize this as

$$U(\vec{\theta}) \approx \prod_k \exp\left(\theta_k [K_k - K_k^\dagger]\right),$$

applying each excitation term in sequence. If the terms do not commute, this product is only an approximation to the true $e^{\sum_k (K_k - K_k^\dagger)}$. However, a remarkable practical finding is that often a single Trotter step is sufficient for UCC within VQE. This is because the variational optimizer can compensate for some Trotter error by adjusting the θ parameters slightly. In other words, the flexibility of the ansatz can absorb the short-time Trotter error, as long as it’s not too large. In many chemical applications, UCCSD implemented as a single Trotterized circuit (with each cluster operator applied once) yielded excellent results compared to exact

energies. Nonetheless, it is important to ensure that the Trotter error remains within acceptable limits, especially if the parameters θ_k become large (which could happen for strongly correlated systems). If needed, one can include multiple Trotter steps (effectively repeating the circuit pattern multiple times with fractionally scaled parameters) to reduce the error at the cost of a deeper circuit.

Trotter Error Bounds: For a qualitative estimate, suppose our cluster operators K_k each have norm of order θ and commutators of order c . Then a single-step $U_{\text{trotter}} = \prod_k e^{\theta(K_k - K_k^\dagger)}$ will differ from the ideal U by terms $\sim \frac{1}{2} \sum_{k < l} \theta^2 [K_k, K_l]$ from the BCH expansion. The energy error in first order can be bounded by $\Delta E \lesssim \sum_{k < l} |\theta_k \theta_l| |[[K_k, K_l]]|$ times some state-dependent factor. If this error is too large, increasing to r trotter steps should reduce it roughly by $1/r^2$ for energy (since the unitary error is linear in $1/r$ and energy is quadratic in state error). In practice, for our pairing Hamiltonian example below, we will find that one trotter step is essentially exact within the precision of our simulation, confirming that first-order trotterization is adequate for the UCC ansatz in this case.

To summarize this section: we break the UCC unitary into a product of simpler unitaries, each acting on a few qubits, using Trotter's formula. The BCH expansion provides insight into the approximation error, and we ensure that this error is small (either by it being naturally small for our system, or by increasing trotter steps if necessary). In the next section, we describe how each factor $\exp(\theta(K - K^\dagger))$ is mapped to an actual quantum circuit of one- and two-qubit gates.

C. UCC Ansatz in the VQE Framework

The Unitary Coupled Cluster ansatz is widely employed as a trial state in the Variational Quantum Eigensolver (VQE) for molecular ground states.

In VQE, one constructs a parameterized unitary circuit $U(\vec{\theta})$, prepares the state $|\Psi(\vec{\theta})\rangle = U(\vec{\theta})|\Phi_0\rangle$, and measures its energy $E(\vec{\theta}) = \langle \Psi(\vec{\theta}) | H | \Psi(\vec{\theta}) \rangle$. A classical optimizer then adjusts the parameters $\vec{\theta}$ to minimize $E(\vec{\theta})$.

Formally, one solves

$$E_{\min} = \min_{\vec{\theta}} \langle \Phi_0 | U^\dagger(\vec{\theta}) H U(\vec{\theta}) | \Phi_0 \rangle,$$

and the optimal state is $|\Psi_{\text{opt}}\rangle = U(\vec{\theta}^*)|\Phi_0\rangle$.

In the case of a UCCSD ansatz,

$$U(\vec{\theta}) = e^{\hat{T}(\vec{\theta}) - \hat{T}^\dagger(\vec{\theta})}$$

ensures $|\Psi_{\text{opt}}\rangle$ is a unitary-rotated HF state. Crucially, U_{UCCSD} must include gates that prepare the reference HF state if it does not coincide with the all-zero qubit state.

Implementation steps include: first, perform a classical Hartree-Fock calculation to obtain molecular orbitals and the one- and two-electron integrals. Then build the second-quantized Hamiltonian $H = \sum_{pq} h_{pq} a_p^\dagger a_q + \frac{1}{2} \sum_{pqrs} h_{pqrs} a_p^\dagger a_q^\dagger a_r a_s$ in that basis. Also define the excitation operators $\hat{\tau}_i$ corresponding to chosen single and double excitations. Next, map the fermionic operators to qubit Pauli operators (e.g. via Jordan-Wigner or Bravyi-Kitaev transforms)

For example, $a_p^\dagger a_q - a_q^\dagger a_p$ maps to a sum of Pauli strings. The mapped UCCSD unitary is then decomposed into native gates using the Trotter formulas above. Each factor $\exp[\theta_i(\hat{\tau}_i - \hat{\tau}_i^\dagger)]$ can be further decomposed into a sequence of single- and two-qubit rotations and CNOTs using standard techniques

During the VQE loop, the parameter set $\vec{\theta}$ (initially set e.g. to MP2 amplitudes or zeros) is optimized. Each iteration involves: (1) preparing the quantum circuit $U(\vec{\theta})$ on the qubits starting from $|\Phi_0\rangle$, (2) measuring expectation values of the Hamiltonian terms (requiring repeated state preparations), and (3) updating $\vec{\theta}$ on a classical computer according to a chosen optimization algorithm. The process repeats until convergence of the energy. Because the ansatz is unitary, the Rayleigh-Ritz variational principle guarantees the energy estimate is an upper bound to the true ground state. In practice, optimization may be performed using gradient-based or gradient-free methods, often requiring many quantum circuit evaluations. The quality of the result depends on the expressiveness of the ansatz and the optimizer's ability to navigate potential local minima

Each term in T corresponds to exciting a specific set of orbitals. On a quantum computer, we first map the fermionic creation/annihilation operators to qubit operators. Common mappings include Jordan-Wigner (JW) and Bravyi-Kitaev, which represent fermionic Fock states as qubit basis states. In JW mapping, for example, each orbital p (which can be either occupied or empty) is mapped to a qubit state $|1\rangle_p$ or $|0\rangle_p$, and an operator a_p (fermion annihilation) is represented by a Pauli string:

$$a_p \rightarrow \left(\prod_{q < p} Z_q \right) \sigma_p^-, \quad a_p^\dagger \rightarrow \left(\prod_{q < p} Z_q \right) \sigma_p^+,$$

with $\sigma^- = \frac{X+iY}{2}$, $\sigma^+ = \frac{X-iY}{2}$. The string of Z (phase factors) encodes the anti-commutation relations. While these phase factors complicate multi-fermion operators, one can often simplify the implementation by choosing a convenient ordering or by working in a subspace that respects certain symmetries (like fixed particle number).

Focus on a single excitation operator: $K_{i \rightarrow a} = a_a^\dagger a_i$ (which replaces a fermion in orbital i with one in orbital a). The anti-Hermitian combination is $K_{i \rightarrow a} - K_{i \rightarrow a}^\dagger = a_a^\dagger a_i - a_i^\dagger a_a$. Mapped to qubits, this becomes an operator that flips qubit i from 1 to 0 and qubit a from 0 to 1 (and the reverse) simultaneously. In fact, one can show that in the subspace of states where qubits i and a have

either $(n_i = 1, n_a = 0)$ or $(n_i = 0, n_a = 1)$, the operator $a_a^\dagger a_i + a_i^\dagger a_a$ acts exactly like an $X_i X_a + Y_i Y_a$ coupling (which is equivalent to an iSWAP interaction). More formally:

$$a_a^\dagger a_i + a_i^\dagger a_a \equiv \frac{1}{2}(X_i X_a + Y_i Y_a),$$

up to a known phase convention. The associated unitary is then:

$$e^{\frac{\theta}{2}(X_i X_a + Y_i Y_a)},$$

which is recognized as a two-qubit Givens rotation that mixes the $|10\rangle_{i,a}$ and $|01\rangle_{i,a}$ states by an angle θ while leaving $|00\rangle$ and $|11\rangle$ (both occupied or both empty) invariant. Indeed, the action on that two-dimensional subspace is exactly a rotation matrix. Many authors call this an $XX + YY$ or iSWAP-like gate. It can be implemented with standard gates: for example, using 2 CNOTs and one single-qubit rotation (in fact, an $XX + YY$ rotation can be decomposed into an R_{XX} or R_{YY} gate which itself can be built from CNOTs and an R_z). If the hardware natively supports an iSWAP gate, one can directly use it to enact such rotations. In the trapped-ion experiment of Ref. , for instance, all-to-all connectivity allowed implementing each pair excitation as a single Givens rotation on the corresponding qubits without additional swap gates.

Circuit Depth Considerations: For UCCSD in a system of M orbitals and N particles, there are $O(N^2 M^2)$ possible double excitations in the worst case. Implementing each as a gate would be prohibitively costly for large systems. Fortunately, in chemistry and also in our pairing model, many of those amplitudes are zero or negligible due to symmetry or the structure of H . One can therefore compile a UCC ansatz by including only the excitations that appear with non-zero amplitudes in, say, perturbation theory, or by using an iterative selection algorithm (ADAPT-VQE). Even after selecting excitations, one can reduce circuit depth by commuting certain operations past each other if they act on disjoint qubits. For example, an excitation on qubits (i, a) commutes with another on (j, b) if $i, a \cap j, b = \emptyset$. These can be executed in parallel on the quantum circuit. In summary, the gate sequence for a trotterized UCC ansatz consists of repeated blocks, each block containing rotations on pairs of qubits (and possibly on four qubits for double excitations that involve two distinct pairs simultaneously – though in many implementations, a double excitation is broken into two sequential single-pair rotations via an ancilla or additional ordering trick). Recent advances propose optimized compilation strategies for UCC circuits to minimize the CNOT count, which is critical for near-term noisy hardware.

Summary: We decompose the unitary coupled cluster operator into elementary Givens rotations on the qubits. Each such gate is implemented with a small number of one- and two-qubit operations. The overall circuit is the

sequence of these rotations (one per cluster operator in T), possibly repeated for multiple Trotter steps. We now apply this general recipe to the nuclear pairing Hamiltonian, which has a structure that allows further simplifications in the ansatz and circuit.

D. Algorithmic Implementation

The implementation of UCCSD within VQE can be summarized as follows

[H]

VQE with UCCSD Ansatz [1] **Classical precomputation:** Perform Hartree–Fock to obtain integrals and reference state $|\Phi_0\rangle$. Build the second-quantized Hamiltonian $\hat{H}$ and identify excitation operators $\{\hat{\tau}_i\}$ (singles, doubles, ...). Choose a mapping (e.g. Jordan–Wigner) to convert $\{\hat{H}, \hat{\tau}_i\}$ into qubit Pauli operators **Ansatz construction:** Define parameterized unitary $U(\vec{\theta}) = \prod_i e^{\theta_i(\hat{\tau}_i - \hat{\tau}_i^\dagger)}$ (Trotterized UCCSD). Initialize parameters $\vec{\theta}$ (e.g. from MP2 amplitudes or zeros). Prepare state $|\Psi(\vec{\theta})\rangle = U(\vec{\theta})|\Phi_0\rangle$ on the quantum processor. Measure the energy $E(\vec{\theta}) = \langle \Psi(\vec{\theta}) | \hat{H} | \Psi(\vec{\theta}) \rangle$ by sampling (estimating each Pauli term) Use a classical optimizer to update $\vec{\theta} \leftarrow \vec{\theta} - \eta \nabla E(\vec{\theta})$ (or other rule) to lower the energy. convergence of E or $\vec{\theta}$ changes small **Return:** Optimized parameters $\vec{\theta}^*$ and energy $E(\vec{\theta}^*)$ (approximate ground state energy).

This high-level algorithm illustrates the hybrid nature of VQE. In practice, efficient gradient evaluation or reduced-measurement schemes may be employed. The accuracy hinges on including sufficient excitation operators in $U(\vec{\theta})$; for UCCSD one includes all single and double excitations with distinct spin/orbital indices. Diagrammatic or tensor-network derivations are usually not needed on the quantum side since the circuit implicitly handles the entangling structure. The mapping step (line 3) is often implemented by software tools (e.g. OpenFermion, Qiskit) which output a list of Pauli strings. The exponential of each string can then be translated to gates via standard formulas.

The Unitary Coupled Cluster (UCC) method offers a powerful, chemically-inspired ansatz for quantum algorithms. It retains the accuracy and size-extensivity of classical CC (via the exponential parametrization) while making the state preparation unitary for quantum hardware. By employing Trotter–Suzuki decompositions, the UCC operator is factorized into implementable gate sequences. Within VQE, UCCSD provides a systematic, variational approach to molecular eigenstates.

Future improvements include adaptive ordering of terms to reduce circuit depth, higher-order Trotter schemes to lower approximation error, and use of symmetries to prune the ansatz. Overall, UCC exemplifies the synergy of many-body theory with quantum computing, bridging traditional electronic structure methods

and near-term quantum algorithms.

V. RESULTS AND DISCUSSIONS

VI. CONCLUSION